

A complex relationship between moonlight and temperature on the foraging behavior of the Alabama beach mouse

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Abstract. Most animal species forage with risk from an ensemble of predators, wherein some predators are themselves prey to the others. A forager's behavior should reflect a synthesis of the effect of environmental conditions on both marginal predation risk and rates of energy accumulation. Here, a forager's giving-up density [GUD] is analyzed for signs of these complex signals. Specifically, we hypothesized that temperature can reverse the effect of moonlight intensity on Alabama beach mouse (*Peromyscus polionotus ammobates*) GUD because changes in temperature change the ensemble of predators from homeothermic taxa with better vision than the mouse to ectothermic taxa with worse vision than the mouse. We fit several models to GUD measurements taken over a broad range of temperatures and nocturnal luminosities. We obtained strong information-theoretic support for a model that is consistent with our hypothesis of temperature-dependent reversal of the effect of nocturnal illumination on mouse GUD. Signals in GUDs can reveal complex effects of optimal foraging at multiple trophic levels, which is especially informative when direct measurement of taxonomic sources of predation is not feasible.

Key words: Alabama beach mouse; foraging behavior; giving-up density, GUD; moonlight; optimal foraging; *Peromyscus polionotus ammobates*; predator ensembles; temperature effects; trophic web.

INTRODUCTION

Most animal species exist in the middle of a food chain or, more generally, a trophic web. They have numerous species that they prey upon and are, themselves, prey for multiple predators. Thus, any given set of environmental conditions that affects foraging of a focal prey species can also affect its predators' abilities to forage as well. This can be made more complex if predator species respond differently to a given set of conditions and if some of the predators are themselves prey for other predators. The focus of this study is to determine whether a focal prey species is able to adaptively adjust to the effects of environmental conditions when a suite of different predators with different environmental requirements are embedded in a complex food web.

Two environmental conditions that affect both predators and prey are ambient light and temperature. Nocturnal light is well known to depress foraging effort by small mammals (e.g., mice), because it increases the forager's risk of being preyed upon. The ability to use light is important to players on both sides of the

predator–prey interaction (Beauchamp 2007). Where the predator is likely to see the prey before the prey sees the predator, more light probably increases the costs of foraging for the prey and thus depresses its foraging effort. Such is often the case for nocturnal rodents being captured by canids, felids, or owls (e.g., Kotler et al. 1988, Longland and Price 1991). However, for predators that detect and strike their prey from very short range, vision may be more useful to the prey than the predator. Webster and Webster (1971) found that this was the case for kangaroo rats (*Dipodomys merriami*) when evading rattlesnakes (*Crotalus cerastes*). Additionally, the speed of snakes decreases quickly with declining temperatures (Heckrotte 1967), which can have two important effects. First, its probability of making a successful strike on a small rodent also is likely to decline with temperature; second, its slower speed makes the snake more likely to be preyed upon by other predators.

Declining ambient temperature is well known to have a negative effect on foragers due to the higher metabolic costs of maintaining homeostasis (e.g., Kotler et al. 1993, Meyer and Valone 1999, Kilpatrick 2003, Orrock and Danielson 2009, Rieucan et al. 2009). However, temperature can affect the predator community for the same reasons that it affects the prey. Where predators and prey are both homeotherms, the effect of temper-

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ature is likely to be much greater for the smaller species. However, in cases where one species is an ectotherm, the effect of temperature is likely to be dramatic and quite different than for a homeotherm.

The foraging literature contains several examples of a strong interaction between the effects of ambient light and temperature that is suggestive of these sorts of differential responses to changes in these conditions among a suite of predators in a trophic web (Kotler et al. 1993, Meyer and Valone 1999, Orrock and Danielson 2009). Our focal species, the Alabama beach mouse (*Peromyscus polionotus ammobates*), is nocturnal and lives in a simple environment that contains a mixture of relatively sparse vegetation and open sand, which can leave the mice exposed to bright moonlight and cold nighttime temperatures. They are vulnerable to a suite of predators that quite likely respond very differently to these two environmental factors, including relatively large mammalian predators dominated by fox (*Vulpes vulpes*), coyotes (*Canis latrans*), and feral cats (*Felis catus*), avian predators (especially Great Horned Owl *Bubo virginianus*), and a variety of colubrid snakes (e.g., *Coluber constrictor*, *Masticophis flagellum*). These predators undoubtedly respond differently to change in light and temperature. Presumably, when temperatures are high, snakes are likely to be strong predators for these mice, so higher light levels should benefit foraging mice. However, when temperatures are low, the costs of foraging for snakes should be high, and thus they are likely to be less important predators to the mice than mammals and birds. When mammals and birds are the primary predation threat, high ambient light levels should be detrimental to foraging mice because these predators use light better than do mice.

METHODS

As a forager removes food items from a depletable patch, consumption rate declines because the forager spends more time searching for the next item. Eventually consumption rate falls to a point where the instantaneous benefit of continued foraging in the patch no longer exceeds the instantaneous cost. At this point the forager should abandon the patch, leaving behind a measurable density of food, known as the giving-up density, or GUD. Brown (1988, 1992) developed this idea to include foraging costs associated with perceived predation risk, metabolic maintenance, and missed opportunities. If nocturnal illumination helps predators to capture their rodent prey more than it helps rodents to evade predators (Kotler et al. 1988, Longland and Price 1991), then rodent GUD should increase with increasing nocturnal illumination (Fig. 1).

We used six clear plastic boxes as experimental feeders. The length, width, and height of each feeder were 33 cm, 7 cm, and 4 cm, respectively. Two holes 5 cm in diameter were cut into the sides of each feeder to

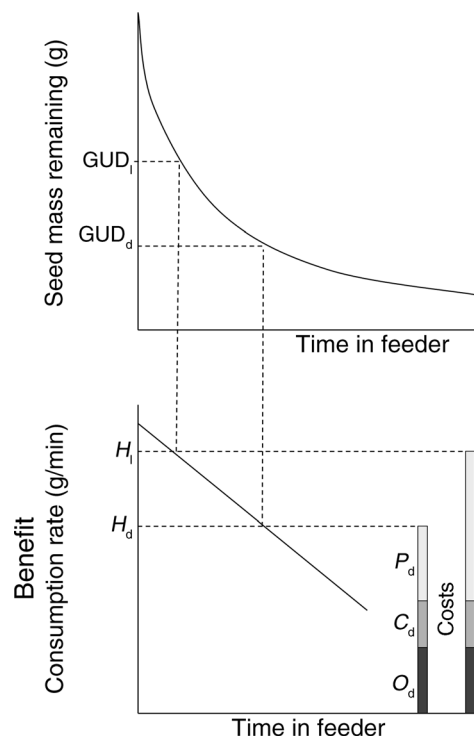


FIG. 1. Theoretical relationship between giving-up density (GUD) for the Alabama beach mouse (*Peromyscus polionotus ammobates*) and the cost of foraging in experimental feeders containing a mixture of sand and seeds. When a forager begins harvesting seeds from a feeder, seed mass is high (upper left portion of upper graph). The density of seeds is also high, so there is a high benefit (consumption rate) of foraging (upper left portion of lower graph). Continued consumption suppresses seed mass and seed density, which produces a concomitant reduction in consumption rate. Eventually, consumption rate falls to a point where the instantaneous benefit of feeding equals the instantaneous costs of feeding (lower right y-axis), and the forager should abandon the feeder. If a nocturnal forager perceives higher predation risk on lighter (P_i) nights than darker (P_d) nights, then a higher consumption rate (H_i) is required to make foraging worthwhile. This higher consumption rate on light nights is associated with a higher giving-up density (GUD_i) in an experimental feeder. "C" and "O" represent metabolic costs and missed opportunity costs, respectively, which are assumed here to be unaffected by nocturnal light intensity.

allow passage of foraging beach mice. We mixed 3 g of millet seeds into 1 L of sand and poured them into each feeder. Feeders were placed in the Ft. Morgan Unit of the Bon Secour National Wildlife Refuge, Gulf Shores, Alabama, USA, in a single row with ~90 m between consecutive feeders. This area is unaffected by anthropogenic light. Extensive live-trapping surveys in this area indicate that the beach mouse is the only rodent species likely to encounter the feeders.

Millet seeds were recovered from feeders that had been left out overnight by pouring the contents of the feeder through a seed-impermeable screen. Feeders were

TABLE 1. Candidate models of the natural log of giving up density (GUD) for the Alabama beach mouse (*Peromyscus polionotus ammobates*).

Model	df	AIC
$\ln(\text{GUD}) = \beta_0 + \beta_1\text{mlx} + \beta_2\text{Kelvin} + \beta_3\text{Kelvin}^2 + \beta_4\text{Kelvin} \times \text{mlx}$	6	-30.4
$\ln(\text{GUD}) = \beta_0 + \beta_1\text{mlx} + \beta_2\text{Kelvin} + \beta_3\text{Kelvin}^2$	5	-23.5
$\ln(\text{GUD}) = \beta_0 + \beta_1\text{mlx} + \beta_2\text{Kelvin} + \beta_3\text{Kelvin} \times \text{mlx}$	5	-18.6
$\ln(\text{GUD}) = \beta_0 + \beta_1\text{mlx}$	3	-17.2
$\ln(\text{GUD}) = \beta_0 + \beta_1\text{mlx} + \beta_2\text{Kelvin}$	4	-16.2
$\ln(\text{GUD}) = \beta_0 + \beta_1\text{Kelvin}$	3	-11.0

refilled with 1 L of sand and 3 g of fresh millet, thoroughly mixed. All matter collected in the screen was taken back to a lab, where millet seeds were separated from other particles (mostly feces), dried in an oven, and then weighed. The mass in grams of the dried seeds is the GUD for a given feeder on a given night. This process was repeated almost every day from 6 February 2007 through 22 May 2007. If the mass of a dried sample was at least 2.7 g, then it was determined that the feeder was not visited and the sample was not included in subsequent analyses. In total, 461 GUD measurements were recorded.

Nightly nocturnal light intensity was measured with a scotopic light sensor (PMA2131) and data logger (PMA2100, Solar Light Company, Glenside, Pennsylvania, USA) placed near the feeders. The device recorded light intensity to the nearest 0.1 millilux (mlx) every five minutes. The average of the recordings taken from half an hour after sunset to half an hour before sunrise was used in subsequent analyses. We also used daily minimum temperatures recorded at Gulf Shores, Alabama, obtained from the Weather Underground website.

We performed a Mantel test on the geographic distance among feeders and the daily correlation of GUD among feeders in order to assess the appropriateness of treating GUDs from each feeder on a given day as spatially independent samples. We nonetheless performed analyses using GUD measurements averaged over all feeders from a given day, which produced a total of $N = 84$ GUD measurements for subsequent analyses. We created an a priori set of regression models to explain observed patterns of GUD (Table 1). Some models include a quadratic term for temperature because we hypothesize that there is an optimum value for foraging mice. Of great interest to us is the possibility

that light intensity and temperature interact with respect to foraging conditions. We included this term in our most complex model, as well as in a model without a quadratic term for temperature. We used Akaike's Information Criterion (AIC) to identify the best model. To assess the significance of each term in the best model, the t statistic for a given term in the full model was compared to a distribution of t statistics from data simulated as full-model predictions plus randomly permuted residuals of a model lacking the focal term (Anderson and Legendre 1999).

RESULTS

Results of the Mantel test indicated that there is no evidence for spatial autocorrelation among feeders ($P = 0.35$ based on 9999 permutations), so all 461 GUD measurements could be analyzed without concern for spatial pseudoreplication (Hurlbert 1984). However, for additional protection against spatial subsampling, we also analyzed our GUD data after averaging over all feeders on a given day. We found that the decision to average over feeders did not affect the AIC rankings of our models, and had very minor effects on the parameter estimates of the best model. If GUDs were averaged across feeders, then the adjusted coefficient of determination was much larger (0.24 vs. 0.07), and the P values for the terms in the best model were also larger. However, the significance of each term in the best model was less than 0.05 even when GUDs were averaged (Table 2).

The significantly positive parameter estimate associated with light intensity suggests that GUDs increase with increasing light intensity (Table 2). However, this relationship is conditional because, as predicted, there is a significant interaction between light intensity and temperature. Because models with squared terms and

TABLE 2. Regression results for the model with lowest AIC; significance was assessed with 9999 permutations of reduced-model residuals.

Term	Estimate	SE	t	P
Intercept	166.6	44.46	3.75	0.0003
Kelvin	-1.183	0.3157	-3.75	0.0002
mlx	0.0355	0.0117	3.04	0.0031
Kelvin ²	2.10×10^{-3}	5.60×10^{-4}	3.75	0.0004
Kelvin $\times$ mlx	-1.25×10^{-4}	4.18×10^{-5}	-2.98	0.0050

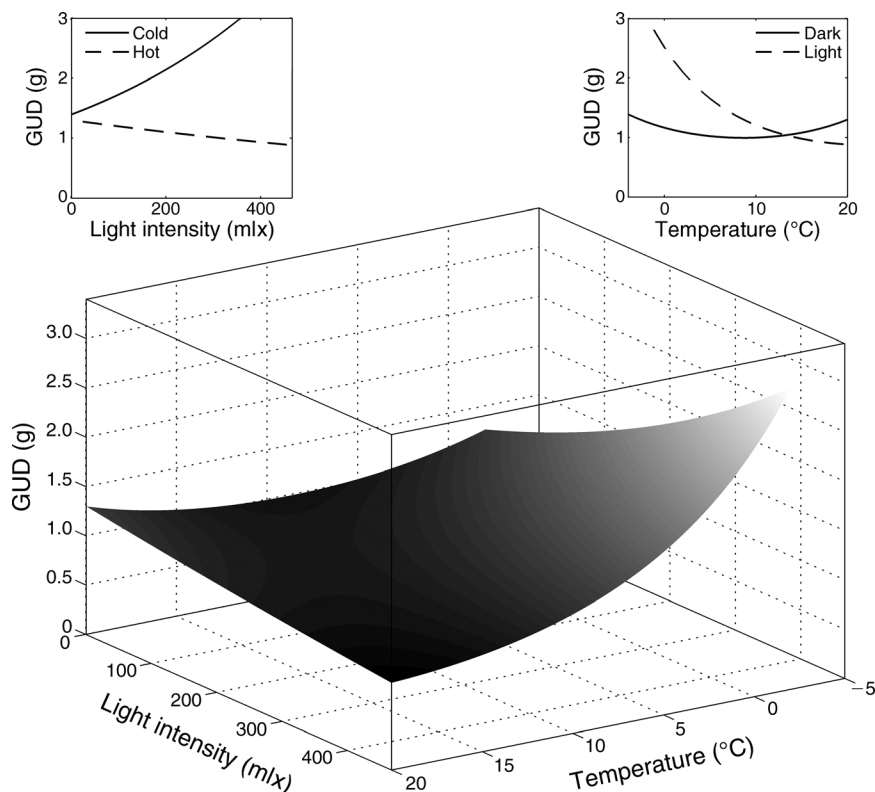


FIG. 2. Predicted giving-up density (GUD) over the observed range of nocturnal light intensity (mlx) and daily minimum temperature. Note that the model was fitted on a Kelvin scale to avoid perceived problems with squaring a negative value. The results were then transformed to the more familiar Celsius scale for the figure.

interactions complicate the interpretation of parameter estimates, the estimates in Table 1 were used to create a surface plot of GUD as a function of light intensity and temperature (Fig. 2).

The surface in Fig. 2 touches the four (transparent) walls of the surrounding box. The four lines representing these points of contact are plotted in the insets of Fig. 2, and display the relationship between GUD and one explanatory variable when the other explanatory variable is at a maximum or minimum value. These lines therefore represent the predicted mouse GUD at observed extremes of light intensity and temperature. GUD increases with nocturnal illumination on relatively cold nights, but decreases with illumination on warm nights. However, GUD is always higher on cold nights than on warm nights, regardless of the level of illumination. On dark nights GUDs have a u-shaped relationship with temperature, but on bright nights, GUDs decline with increasing temperature.

DISCUSSION

Increased visual exposure to predators should increase the cost of foraging and result in increased GUDs. Numerous studies have demonstrated that rodent GUD

increases with nocturnal illumination (Kotler et al. 1991, 1993, 2010, Hughes et al. 1995, Yunker et al. 2002), including our focal species (Bird et al. 2004). Indeed, this is precisely what occurred over most of the range of observed temperatures. However, this relationship reversed when daily minimum temperatures became relatively high. This is consistent with the hypothesis that, as temperature increases, threat of predation increasingly comes from snakes, and the advantage of illumination shifts from predator to prey. Snakes strike from close range, and beach mouse habitat is often sparsely vegetated. This gives mice an opportunity whereby, with sufficient light, they may detect snakes and avoid attack. Webster and Webster (1971) showed experimentally that kangaroo rats strongly benefit from good vision in avoiding rattlesnakes. Interestingly, kangaroo rats and rattlesnakes occupy similarly sparsely vegetated habitat, where vision would appear to be a plausibly useful sensory system for snake avoidance. Additionally, snakes are themselves at risk of prey by some of the other beach mouse predators (especially owls) with good night vision. On cool nights, they will be more vulnerable to being preyed upon simply because they cannot move as quickly in cooler ambient

temperatures (Heckrotte 1967). Less mobility may also reduce the likelihood of a successful strike even if a beach mouse is encountered. These thermal issues increase foraging costs for snakes and thereby restrict their foraging effort at low temperatures, leaving homeothermic predators as the predominant risk of mortality to the foraging beach mouse. Thus, optimally foraging snakes should increase activity on warm, dark nights relative to cool, bright nights (Clarke et al. 1996), and our results suggest that beach mice appear to adjust their foraging efforts accordingly.

For a given amount of illumination, we also note that GUDs on cold nights are always greater than GUDs on warm nights. This suggests that cold temperatures increase the metabolic cost of foraging for mice, which is consistent with other findings (Kotler et al. 1993, Meyer and Valone 1999, Kilpatrick 2003). The effect of temperature on GUD is more pronounced on light nights than on dark nights. On bright nights, beach mice become exposed to visually superior predators, and the threat of predation may be further compounded by the cold because mice enter a mild state of torpor (observed by authors). During such nights, a relatively high consumption rate is required to compensate for the additional cost of foraging while visually and physiologically exposed to predators. The threat abates as temperature rises because mice are presumably more alert and mobile, and can therefore "afford" to continue to feed until food patches become relatively depleted and consumption rate is relatively low.

The feedbacks between individual foraging behaviors and population dynamics can provide a mechanistic understanding of many population-level ecological phenomena (Holt and Kimbrell 2007). Foraging theory also provides mechanisms of species coexistence and predator-prey interactions that can explain community-level ecological phenomena (Kotler and Brown 2007). Here, we not only suggest that there is interplay between two abiotic environmental factors with respect to individual foraging behavior, but we also suggest that the interplay occurs because of changes in the behavior of predators. Predators have their own GUDs. Thus, light and temperature not only have direct effects on mouse predation and metabolic costs, but also affect each predator's GUD, which indirectly affects mouse predation costs.

Although the parameters of the best model are highly statistically significant, the coefficient of determination of the model is very small. Residual variation in GUD could be partially explained by variation in many, as yet unknown, environmental variables. For instance, the forage quality of surrounding habitat can change hourly as winds continuously expose and cover seeds on the sandy landscape (Lortie et al. 2000) that beach mice inhabit. Wind will also affect the predators' ability to hear, see, or smell beach mice. Furthermore, this effect

may be different among the various species of predators and perhaps also may depend on other environmental conditions. For instance, Wells (1978) and Wells and Lehner (1978) found that coyotes strongly defer to vision over any other sense, but use either auditory or olfactory senses as alternative modes of detecting prey in very dark conditions, depending upon wind and possibly distance from the prey. Meanwhile, Great Horned Owls rely on sound to locate prey when vision is not sufficient (Volman and Konishi 1989, Volman 1994). In summary, a host of environmental conditions affect both predator and prey sensory systems and these effects will differentially impact the costs and benefits of foraging for each species. The effect of changing environmental conditions on a focal foraging animal includes indirect effects that ricochet through the food web and ultimately determine how it should best optimize its behavior.

Our results are qualitatively consistent with a change in predator behavior and a differential effect of temperature on metabolic costs across the predator community as well as the prey. The ability to detect these signals in the foraging behavior of a focal species suggests that careful examination of patterns of variability in GUDs can reveal complex ecological processes. Our observational approach circumvents some of the ethical and legal issues associated with manipulative experimentation with species listed under the Endangered Species Act, such as the Alabama beach mouse. Further, wherever logistical constraints prevent the manipulation and measurement of entire communities, the foraging behavior of a focal species can be a useful indicator of broader ecological processes. To better understand the vast variability we see in foraging strategies, future work might profitably search for additional environmental variables that will differentially affect the sensory systems of predators and their prey.

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LITERATURE CITED

- Anderson, M. J., and P. Legendre. 1999. An empirical comparison of permutation methods for tests of partial regression coefficients in a linear model. *Journal of Statistical Computation and Simulation* 62:271–303.
- Beauchamp, G. 2007. Exploring the role of vision in social foraging: what happens to group size, vigilance, spacing, aggression and habitat use in birds and mammals that forage at night? *Biological Reviews* 82:511–525.

- Bird, B. L., L. C. Branch, and D. L. Miller. 2004. Effects of coastal lighting on foraging behavior of beach mice. *Conservation Biology* 18:1435–1439.
- Brown, J. S. 1988. Patch use as an indicator of habitat preference, predation risk and competition. *Behavioral Ecology and Sociobiology* 22:37–47.
- Brown, J. S. 1992. Patch use under predation risk. I. Models and predictions. *Annales Zoologici Fennici* 29:301–309.
- Clarke, J. A., J. T. Chopko, and S. P. Mackessy. 1996. The effect of moonlight on activity patterns of adult and juvenile prairie rattlesnakes (*Crotalus viridis viridis*). *Journal of Herpetology* 30(2):192–197.
- Heckrotte, C. 1967. Relations of body temperature, size, and crawling speed of the common garter snake, *Thamnophis s. sirtalis*. *Copeia* 4:759–763.
- Holt, R. D., and T. Kimbrell. 2007. Foraging and population dynamics. Pages 365–395 in D. W. Stephens, J. S. Brown, and R. C. Ydenberg, editors. *Foraging: behavior and ecology*. University of Chicago Press, Chicago, Illinois, USA.
- Hughes, N. F., D. Ward, and M. R. Perrin. 1995. Effects of substrate on foraging decisions by a Namib desert gerbil. *Journal of Mammalogy* 67:638–645.
- Hurlbert, S. H. 1984. Pseudoreplication and the design of ecological field experiments. *Ecological Monographs* 54:187–211.
- Kilpatrick, A. 2003. The impact of thermoregulatory costs on foraging behaviour: a test with American Crows (*Corvus brachyrhynchos*) and eastern grey squirrels (*Sciurus carolinensis*). *Evolutionary Ecology Research* 5:781–786.
- Kotler, B. P., and J. S. Brown. 2007. Community ecology. Pages 397–434 in D. W. Stephens, J. S. Brown, and R. C. Ydenberg, editors. *Foraging: behavior and ecology*. University of Chicago Press, Chicago, Illinois, USA.
- Kotler, B. P., J. S. Brown, and O. Hasson. 1991. Factors affecting gerbil foraging behavior and rates of owl predation. *Ecology* 72:2249–2260.
- Kotler, B. P., J. S. Brown, and W. A. Mitchell. 1993. Environmental factors affecting patch use in two species of gerbilline rodents. *Journal of Mammalogy* 74:614–620.
- Kotler, B. P., J. S. Brown, S. Mukherjee, O. Berger-Tal, and A. Bouskila. 2010. Moonlight avoidance in gerbils reveals a sophisticated interplay among time allocation, vigilance and state-dependent foraging. *Proceedings of the Royal Society B* 277:1469–1474.
- Kotler, B. P., J. S. Brown, R. J. Smith, and W. O. Wirtz II. 1988. The effects of morphology and body size on rates of owl predation on desert rodents. *Oikos* 53:145–152.
- Longland, W. S., and M. V. Price. 1991. Direct observations of owls and heteromyid rodents: can predation risk explain microhabitat use? *Ecology* 72:2261–2273.
- Lortie, C. J., D. T. Ganey, and B. P. Kotler. 2000. The effects of gerbil foraging on the natural seedbank and consequences on the annual plant community. *Oikos* 90:399–407.
- Meyer, M. D., and T. J. Valone. 1999. Foraging under multiple costs: the importance of predation, energetic, and assessment error costs to a desert forager. *Oikos* 97:571–579.
- Orrock, J. L., and B. J. Danielson. 2009. Temperature and cloud cover, but not predator urine, affect winter foraging of mice. *Ethology* 115:641–648.
- Rieuciau, G., W. L. Vickery, and G. J. Doucet. 2009. A patch use model to separate effects of foraging costs on giving-up densities: an experiment with white-tailed deer (*Odocoileus virginianus*). *Behavioral Ecology and Sociobiology* 63:891–897.
- Volman, S. F. 1994. Directional hearing in owls: neurobiology, behaviour and evolution. Pages 292–314 in M. N. O. Davies and P. R. Green, editors. *Perception and motor control in birds: an ecological approach*. Springer-Verlag, Berlin, Germany.
- Volman, S. F., and M. Konishi. 1989. Spatial selectivity and binaural responses on the inferior colliculus of the great horned owl. *Journal of Neuroscience* 9:3083–3096.
- Webster, D. B., and M. Webster. 1971. Adaptive value of hearing and vision in kangaroo rat predator avoidance. *Brain, Behavior and Evolution* 4:310–322.
- Wells, M. C. 1978. Coyote senses in predation: environmental influences on their relative use. *Behavioural Processes* 3:149–158.
- Wells, M. C., and P. N. Lehner. 1978. The relative importance of the distance senses in coyote predatory behaviour. *Animal Behaviour* 26:251–258.
- Yunger, J. A., P. L. Meserve, and J. R. Gutiérrez. 2002. Small-mammal foraging behavior: mechanisms for coexistence and implications for population dynamics. *Ecological Monographs* 72:561–577.